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RESEARCH ARTICLE

Modelling: effect of feeding plantain pasture to three different breeds of calf/heifer on weight gain and nitrogen leaching in Canterbury

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ABSTRACT

Four levels of plantain (0%, 25%, 50% and 75% on DM basis) inclusion rate to ryegrass-clover pasture and three target heifer mature live weights, 444 kg (representing Jersey), 529 kg (Friesian), and 492 kg (cross bred Jersey × Friesian) were tested in this modelling exercise for a 30 ha farm and 24-month analysis. The annual average growth rates were 30 kg DM/ha/day and 23 kg DM/ha/day for plantain and perennial ryegrass-white clover, respectively. Feeding plantain increased animal live weight gain, but on a whole farm level, total live weight gain was similar between the three target weight treatments. Stocking rate was identified as a major driver of nitrogen leaching in this study, which resulted in a decreased nitrogen leaching with increased target live weight groups. Increased annual nitrogen leaching value was found from feeding plantain to animals, which contrasts to previous speculation from animal studies.

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Nitrate leaching; urine nitrogen; herbal pasture; prediction equation; cattle growth

Introduction

The New Zealand dairy industry is one of the country's largest export earners and dairy calf rearing is a crucial part of the industry as adequate live weight (LW) gain, prior to first mating and calving, is vital to ensure both high reproductive performance and milk production (Macdonald 2005). In New Zealand, dairy calves commonly graze pasture mixtures containing perennial ryegrass (*Lolium perenne L.*) and white clover (*Trifolium repens L.*). While this pasture mixture has many desirable characteristics (Kenyon et al. 2010), it may suffer from low growth rates in summer and autumn and also has variable nutritive value in spring and summer (Valentine and Kemp 2007; Parsons et al. 2011), which may lead to low nutrient intake by grazing calves and limit their growth (Burke et al. 2002; Muir et al. 2014).

In recent years, there has been increased interest in the inclusion of forage herbs such as plantain (*Plantago lanceolata L.*) into grazed pastures to maintain feed forage quality and production throughout the year as well as increase animal intake and performance (Schreurs et al. 2002; Chapman et al. 2008; Muir et al. 2014). Despite previous studies

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demonstrating higher LW gain performance in sheep and deer grazing swards containing plantain, compared to perennial ryegrass-white clover pastures (Schreurs et al. 2002; Kenyon et al. 2010), there have been few studies into the effects of plantain inclusion rate on calf LW gain (Cheng et al. 2017b; Cheng et al. 2017c). Further, differences in the genetic background of animals, such as breeds, are also known to be a major factor influencing LW gain. Therefore the first objective of this study was to investigate the effects of including four levels of plantain in perennial ryegrass-white clover pasture with three dairy calf breeds on LW gain, using a feed profiling model based on the principles and equations presented in Webby and Bywater (2007) and as described in a previous study (Gregory 2015), with seasonal plantain and pasture production data collected from a field study in Canterbury, New Zealand (Martin et al. 2017).

Recent studies showed pastures containing plantain potentially offer a strategy to reduce environmental impacts of livestock farming (Pembleton et al. 2015; Cheng et al. 2017a; Cheng et al. 2017b). For example, lactating cows grazing a pasture mixture containing 18% plantain had lower urinary nitrogen excretion (UN) and urinary nitrogen (N) concentration than cows grazing perennial ryegrass-white clover pasture (Totty et al. 2013). Cheng et al. (2017a) and Cheng et al. (2017b) showed dairy heifers had lower UN and urinary N concentration when offered plantain, compared to when they were offered pasture. However, to the best of our knowledge, no study has investigated how much plantain is needed in the diet to achieve a reduction in UN and urinary N concentration from calves, or the potential impact of a change in UN and urinary N concentration on N leaching in the soil on a paddock or farm scale. Therefore, the second objective of this study was to assess the interactive effects of including four levels of plantain in pasture-based diet with three different dairy calf breeds, on UN and urinary N concentration, and N leaching using established prediction equations generated from our previous studies.

Materials and methods

Model

The model analysis was carried out using a modified feed profiling model (Gregory 2015, Webby and Bywater 2007), which balanced feed supply and demand in half monthly periods over one year for 30 ha land. Pasture growth rates, metabolisable energy (ME), N concentration in the pasture and utilisation rates were specified by the user. Calf energy requirement for maintenance and growth were calculated using AFRC (1995), based on user specified animal starting LW and growth rates.

Pasture dataset

The pasture dataset used for model development was collected from 8th October 2014 to 30th August 2016 at the Lincoln University Research Dairy Farm, Canterbury, New Zealand (43°64′S, 172°46′E) on a free-draining Templeton fine sandy loam (Immature Pallic soil; Hewitt 2010).

The experiment consisted of a plot trial using monocultures of plantain (*cv.* Tonic), perennial ryegrass (*cv.* One50 AR37) and white clover (*cv.* Kopu II). Plots were 3 m × 2.1 m in size and replicated 4 times to reduce environmental variation. The pastures were fertilised with a rate of 180 kg N/ha/year for perennial ryegrass and plantain, and 156 kg N/ha/year for white clover. Fertiliser was applied in split applications after each harvest in compliance with DairyNZ guideline (Dairy 2016). The area was irrigated with a travelling irrigator between October and March yearly with approximately 20–30 mm water applied per week (total 550 mm).

The site was managed according to best practice methods (Moot et al. 2003; Lee et al. 2011; Dairy 2013) with a cut and carry approach, which resulted in no animals grazing the plots. Pasture was harvested nine times per year for perennial ryegrass and plantain and seven times per year for white clover. This allowed perennial ryegrass to reach third leaf stage (Donaghy and Fulkerson 1998) and mimicked a typical rotation for a perennial ryegrass-white clover pasture. At each harvest, one 3 m × 0.45 m strip from each plot was cut with a Briggs & Stratton 650 series 190cc rotary blade push mower (Milwaukee, USA) to a height of 4 cm and the herbage was collected in the catcher. The herbage collected was weighed to determine fresh weight and two subsamples were taken. The first subsample was oven-dried at 60°C for 48 h to determine dry matter percentage (DM%) (Adesoganl et al. 2000) and multiplied by fresh weight to determine herbage yield. Growth rates for each species were calculated using these yield values. The second subsample was frozen and freeze dried before being ground through a 1 mm sieve using a M200 rotor mill (Retsch Inc. Newtown, Pennsylvania, USA) and scanned by calibrated near infra-red spectrophotometer (NIRS, NIRSystems 5000, Foss, Maryland, USA) to determine N (% of DM) and metabolisable energy (ME; MJ/kg DM). For each pasture species; growth rate, N% and ME were averaged over half monthly period before being loaded into the model.

A pasture renewal programme was also included in the model with 10% of the farm being regrassed annually according to the suggestion of Glassey et al. (2010). Regrassing occurred in spring with a 20% increase in new growth from the new pasture every two weeks for the first ten weeks of regrassing.

Animal data

The model included a total period of 24 months from July to June with calves brought on to the farm at 3 months of age (i.e. post weaning) in November and removed from the farm at 24 months old in August. The LW gains were estimated from target LW data according to DairyNZ (2015) recommendations, for three commonly used dairy breeds on NZ dairy farms: Jersey (target mature LW 444 kg), Friesian (target mature LW 529 kg), and cross bred (Jersey × Friesian, target mature LW 492 kg) calves/heifers with seasonal growth rates based on targets recommended by the NZ Beef council (MAF, date unknown). Weaning weight was assumed as 18.9% of end target LWs. To simplify the analysis and make comparison between scenarios easier, the mortality rate was set at 0%, effectively meaning that any deaths would have occurred before calves were weaned and brought on the farm.

Scenarios

Growth rate, N% and ME for plantain were taken directly from the pasture data set. A perennial ryegrass-white clover mixture was developed using a ratio of 85% perennial ryegrass: 15% white clover (DM basis), with data formulated from the original plots measurements. Four proportions of plantain and ryegrass-white clover mixture were established to represent a range of mixtures in the dairy farm system:

100% perennial ryegrass-white clover: 0% plantain

75% perennial ryegrass-white clover: 25% plantain

50% perennial ryegrass-white clover: 50% plantain

25% perennial ryegrass-white clover: 75% plantain

The three target LW profiles and four different pasture mixtures of ryegrass-white clover: plantain generated 12 different scenarios shown in Table 1.

Operating rules for the model

For each scenario, stock numbers and the amount of pasture conserved as silage and fed back to the animals were adjusted to balance feed demand and supply over the year. To

Table 1. Feed demand and supply at different heifer target LW and different proportions of perennial ryegrass-white clover and plantain.

Scenario	Target LW (kg)	Perennial ryegrass-white clover (%)	Plantain (%)	Total number animals on farm	Feed supply (T DM/ha/year)	Feed demand (T DM/ha/year)	Supplement made (T DM/year)	Supplement fed (T DM/year)	Surplus (T DM/year)
1	444	100	0	170	8.0	7.6	7.4	7.3	0.48
2	444	75	25	183	8.6	8.2	8.2	8.2	0.03
3	444	50	50	195	9.2	8.8	8.7	8.6	0.41
4	444	25	75	208	9.8	9.4	9.1	9.1	0.04
5	492	100	0	153	8.0	7.6	7.6	7.5	0.27
6	492	75	25	164	8.6	8.2	8.0	8.0	0.31
7	492	50	50	176	9.2	8.8	8.3	8.3	0.16
8	492	25	75	187	9.8	9.4	9.2	9.2	0.47
9	529	100	0	142	8.0	7.7	7.5	7.5	0.01
10	529	75	25	152	8.6	8.2	7.9	7.9	0.04
11	529	50	50	162	9.2	8.8	8.5	8.5	0.21
12	529	25	75	172	9.8	9.3	9.3	9.3	0.02

ensure the model is representative of the best farm practice and that results are comparable across scenarios, a number of operating rules were established:

- Starting cover at the first of July was set at 1950kg DM/ha for perennial ryegrass–white clover, calculated using a target post-grazing mass of 1500 kg DM/ha and a target pre-grazing mass of 2400 kg DM/ha. For plantain, the starting cover at first of July was set to 1675 kg DM/ha, based on a post-grazing mass of 800 kg DM/ha and a pre-grazing mass of 2550 kg DM/ha. The lower post-grazing mass in plantain was a reflection of a lower DM% compared to perennial ryegrass–white clover with cutting heights being the same, when harvesting both herbages. Overall starting cover varied between each scenario depending on the proportion of plantain and perennial ryegrass-white clover in the farm system.
- End cover in June must be within 50 kg DM/ha of starting cover to ensure the system is in 'steady-state'.
- To ensure the system is in balance, average cover throughout the year must differ by no more than 100 kg DM/ha from the farm starting cover at the first of July. This ensured post-grazing mass was not grazed too low to cause depleted plant energy reserves and low annual growth rates, or pre-grazing mass was too high to cause a lower feed quality (Fulkerson and Donaghy 2001).
- All silage was made on the farm, no silage deficit could occur and no more than 0.5 t (T) DM surplus could be left per year; this allowed the farm system to be self-sustaining.
- No more than 50% of the farm was cut for silage. This, and the previous rules were intended to ensure that the farm operated as a pastoral grazing system and not a cut and carry operation. Conservation of surplus pasture and feeding as silage, was necessary to match what was essentially a linear demand curve with a seasonal pasture supply pattern.
- The primary outputs from the model were the number of stock carried and expected N consumption and excretion per ha. Total feed supply and demand, starting and ending covers and silage balance were used to ensure each scenario followed the same operating principles.

Estimation of urinary N excretion and urinary N concentration

The UN and urinary N concentration were estimated based on N intake. The estimation equations were established using data from previous studies on dairy heifers grazing on plantain/ryegrass-white clover pasture in Canterbury under irrigation (Cheng et al., unpublished). The studies comprised four grazing trials and one indoor trial, and heifers were offered different proportions of plantain and perennial ryegrass-white clover in their diet. Equations derived from these data were:

For 100% and 75% perennial ryegrass-white clover treatments:

Urinary N concentration (%) = 0.001466 × N intake (g/day/animal) − 0.0143 ($n = 10$; $R^2 = 82.1$; $P < 0.001$)

UN (g/day/animal) = 0.3512 × N intake (g/day/animal) + 34.72 ($n = 11$; $R^2 = 85.7$; $P < 0.001$)

For 50% plantain and 75% plantain treatments:

Urinary N concentration (%) = 0.000940 × N intake (g/day/animal) + 0.0715 ($n = 9$; $R^2 = 64.5$; $P = 0.006$)

UN (g/day/animal) = 0.391× N intake (g/day/animal) + 27.8 ($n = 9$; $R^2 = 59.7$; $P = 0.009$)

The above equations are in general agreement with previous work in dairy cattle (Dewhurst et al. 2010; Cheng et al. 2011), who demonstrated the interrelationships between UN, urinary N concentration and N intake.

Estimation of urinary N loading and leaching levels

Total seasonal UN loading for the 30 ha block (UN_L) was calculated in the following manner;

$UN_L = ((H_n \times UN_{ex})/1000) \times 91$ days

Where H_n = animal numbers for the specific treatment, UN_{ex} = mean daily UN excreted (g/day/animal for that treatment).

Seasonal urine patch numbers 30 ha^{-1} ($UP_n^{\dagger}$) was calculated as follows;

$UP_n = H_n \times 10$ urinations/day$^{\dagger} \times 91$ days

Mean N loading within urine patches (UPN_L) was calculated as follows;

$UPN_L = ((UN_L/UP_n)/0.2\ m^2$ patch surface area$^{\dagger}) \times 10000\ m^2$ [1 ha]

Where † assumptions and values were derived from Moir et al. (2011).

N leaching loss per urine patch (UNLP) was calculated using the equation from Figure 3 of Di and Cameron (2007, NCAE) and modifying spring and summer leaching by a factor of 10% and 20% respectively, based on the work of Carlton et al. (2016).

N leached (NL; kg N/ha/season) was calculated as;

NL_S = (UNLP/((10,000 m^2/ha/ 0.2 m^2 patch surface area) × UP_n))/30 ha

Total annual N leached (NL_T; kg N/ha/year) was then calculated by summing NL_S for all four seasons.

Results

Forage growth and chemical composition

The annual average growth rates were 30 kg DM/ha/day and 23 kg DM/ha/day for plantain and perennial ryegrass-white clover, respectively (Figure 1a). Plantain growth was consistently higher than perennial ryegrass-white clover between September and April (Figure 1a).

The annual average ME was similar between plantain and perennial ryegrass-white clover (12.2 MJ/kg DM and 12.3 MJ/kg DM, respectively, Figure 1b). However, this varied throughout the year. Between September and December (spring/early summer), plantain had a lower ME than perennial ryegrass-white clover (11.6 vs. 12.4 MJ/kg DM, respectively). On the other hand, between March and April (autumn) the opposite occured (11.5 MJ/kg DM for perennial ryegrass-white clover and 12.5 MJ/kg DM for plantain, respectively, Figure 1b).

The average N concentration between plantain and perennial ryegrass-white clover was similar across the year (Figure 1c). However, there were differences among months. Between August and October (later winter/spring), N concentration in plantain was higher than perennial ryegrass-white clover (3.1% vs. 2.8%, respectively, Figure 1c), while between December and March, plantain had a lower N concentration than perennial ryegrass-white clover (2.4% vs. 2.6%, respectively, Figure 1c).

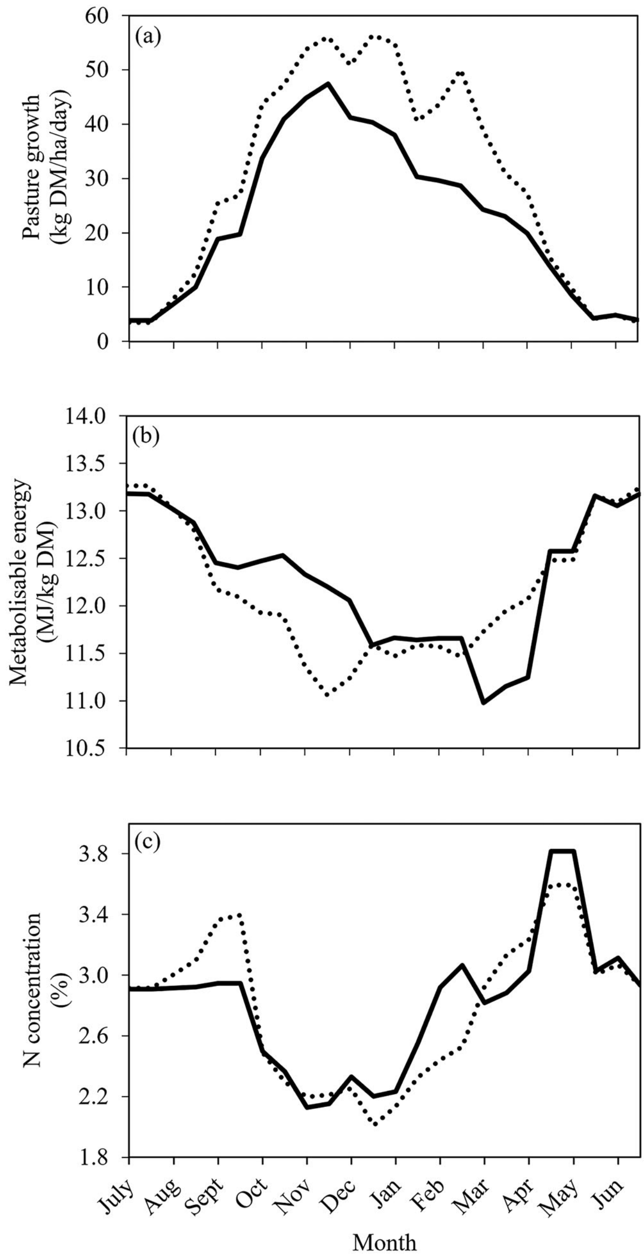

Figure 1. Plot mowing pasture growth rates **a**, Metabolisable energy **b** and N concentration **c** for perennial ryegrass–white clover (—) and plantain (•••) in Canterbury under irrigation.

Stock numbers

Senarios with higher target LW had lower total stock number on farm compared to those with lower target LW on the same forage treatment (Table 1). Within the same target LW, as plantain became a larger part of the diet, more animal was carried on the farm (Table 1).

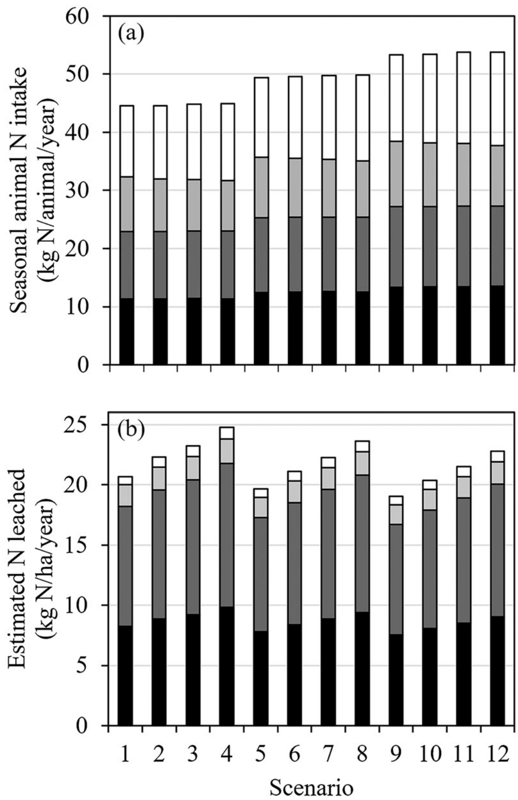

Figure 2. Seasonal heifer N intake **a** and total N leaching estimated **b** in winter (■), autumn (■), summer (□) and spring (□) at each scenario.

Animal N intake and LW gain

The effects of forage and target LW on animal N intake and LW gain are presented in Table 2. Across all scenarios, on a seasonal basis, animal N intake was the highest in spring (14.1 kg N/animal) and the lowest in summer (10 kg N/animal, Figure 2a). The seasonal N intake did not differ between forage type in the autumn and winter (Figure 2a). However, in spring, as the plantain proportion increased, the seasonal N intake increased from 13.5 kg N/animal to 14.7 kg N/animal (Figure 2a). In contrast in summer, as the plantain proportion increased, the seasonal N intake declined from 10.4 kg N/animal/season to 9.6 kg N/animal (Figure 2a).

On a total farm basis, N intake (kg N/ha/year) and total LW gain (kg LW gain/ha/year) were similar between the three target LW treatments (Table 2). Averaged over all the forage treatments, total N intake ranged from 243 kg N/ha/year (for 444 kg target LW) to 241 kg N/ha/year (for 529 kg target LW) and total LW gain ranged from 1147 kg LW gain/ha/year (for 444 kg target LW) to 1134 kg LW gain/ha/year (for 529 kg target LW) (Table 2).

Between the forage treatments, as proportion of plantain increased, total farm N intake increased from 218 kg N/ha/year (for 0% plantain) to 267 kg N/ha/year (for 75% plantain). In addition to this, as the proportion of plantain increased, total LW gain/ha/year

Table 2. Animal N intake and LW gain at different target LW and different proportions of perennial ryegrass-white clover and plantain.

			Seasonal N intake (kg/animal/season)					
Scenario	Target LW (kg)	Plantain (%)	Spring	Summer	Autumn	Winter	N intake (kg N/ha/year)	Total LW gain (kg LW gain/ha/year)
1	444	0	12.2	9.4	11.6	11.3	219	1031
2	444	25	12.6	9.1	11.5	11.3	235	1110
3	444	50	12.9	8.9	11.6	11.4	251	1183
4	444	75	13.2	8.7	11.6	11.3	268	1262
5	492	0	13.6	10.4	12.8	12.4	218	1028
6	492	25	14.1	10.1	12.9	12.5	234	1102
7	492	50	14.5	9.9	12.8	12.6	251	1183
8	492	75	14.8	9.7	12.8	12.6	267	1257
9	529	0	14.8	11.3	13.8	13.3	218	1026
10	529	25	15.3	11.0	13.8	13.4	234	1098
11	529	50	15.7	10.7	13.9	13.4	250	1170
12	529	75	16.1	10.5	13.8	13.5	265	1243

increased from 1028 kg/ha/year (for 0% plantain) to 1254 kg/ha/year (for 75% plantain) (Table 2).

N excretion and leaching potential

The effects of forage treatments and target LW on UN and N leaching potential are presented in Table 3. Averaged over all forage mixtures, animal UN increased as the target LW increased from 76.8 g N/animal/day (for 444 kg target LW) to 85.8 g N/animal/day (for 529 kg target LW). No difference was found in UN per animal between forage treatments (Table 3).

Total N leaching levels were lower in spring and summer (0.8 and 1.8 kg N/ha/season, respectively) than in autumn and winter (10.5 and 8.6 kg N/ha/season, respectively) (Figure 2b, Table 3). Averaged over the forage treatments, N leaching decreased as target LW increased (22.7 vs. 20.9 kg N/ha/year, Table 3). Between forage treatments, the annual N leaching value increased when the proportion of plantain in the diet was higher (19.8 vs. 23.7 kg N/ha/year, Figure 3a, Table 3).

Table 3. Animal N excretion and leaching potential (30 ha) at different target LW and different proportions of perennial ryegrass-white clover and plantain.

Scenario	Target live-weight (kg)	Plantain (%)	Urinary N excretion (g/animal/day)	N in urine (%)	N leaching (kg/ha/year)	Total N eaten (g N/kg LW gain)	Total N excreted (g N/kg LW gain)	Total N leached (g N/kg LW gain)
1	444	0	77.7	0.169	20.7	211.9	136.3	20.0
2	444	25	77.7	0.169	22.3	211.5	136.4	20.1
3	444	50	75.9	0.182	23.2	212.0	133.1	19.6
4	444	75	76.0	0.183	24.8	212.2	133.4	19.6
5	492	0	82.3	0.189	19.7	212.0	130.4	19.1
6	492	25	82.5	0.190	21.1	212.2	130.7	19.1
7	492	50	81.2	0.195	22.3	212.5	128.6	18.8
8	492	75	81.3	0.195	23.6	212.4	128.8	18.8
9	529	0	86.1	0.205	19.0	212.8	126.9	18.6
10	529	25	86.2	0.206	20.4	212.6	127.1	18.5
11	529	50	85.5	0.204	21.5	213.4	125.9	18.4
12	529	75	85.6	0.205	22.8	213.2	126.1	18.3

The N leaching increased as number of animals on the farm increased (Figure 3b). For example, averaged over the pasture mixtures, the 444 kg target LW farms had an average of 189 animals and leached an average of 22.7 kg N/ha/year. In contrast, the 529 kg target LW farms had an average of 157 animals and leaching an average of 20.9 kg N/ha/year. Furthermore, when comparing N leaching levels at different annual N intakes, on a total farm basis (kg N/ha/year), the results show N leaching increased as N intake increased from 20.8 kg N/ha/year to 27.4 kg N/ha/year (Figure 3c).

From N use efficiency point of view, where N equates to g N/ha/year and LW gain equates to kg LW/ha/year, total N intakes were marginally higher per kg LW gain in the 529 target LW animals (211.9 g N eaten/kg LW gain) compared to the 444 target LW animals (213.0 g N eaten/kg LW gain, Table 3). When comparing between the pasture mixtures, animals on the high plantain % pastures had a similar N intakes compared to the low plantain % pastures (212.2 vs. 212.6 g N eaten/kg LW gain, Table 3). In contrast, total N excretion was higher per kg LW gain in the 444 target LW animals (134.8 g N excreted/kg LW gained) compared to the 529 target LW animals (126.5 g N excreted/ kg LW gain, Table 3). When comparing between the pasture mixtures, animals on the high plantain % pastures excreted less N per ha per year compared to the 100% perennial ryegrass-white clover pastures, however not by a large amount

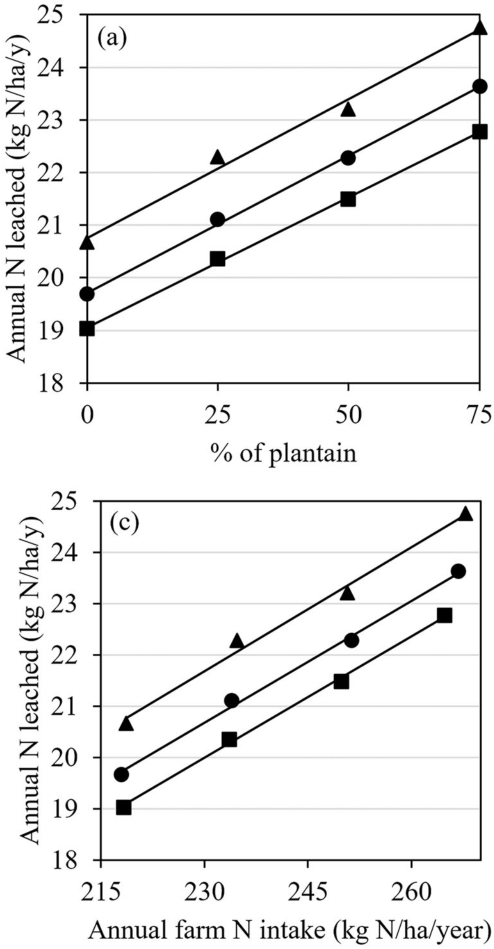

Figure 3. Total N leached (30 ha) with increasing proportion of plantain in the sward **a**, increasing total number of animals **b**, increasing annual farm N intakes (c) and increasing total LW gain (d) at three different target LWs: 444 kg (—▲—), 492 kg (—●—) and 529 (—■—).

(131.2 vs. 129.4 g N excreted/kg LW gain, Table 3). Total N leaching per kg LW gain was higher in the 444 target LW animals (19.8 g N leached/kg LW gain, Table 3) compared to the 529 target LW animals (18.4 g N leached/kg LW gain, Table 3, Figure 3d). When comparing between the pasture mixtures, animals on the high plantain % pastures excreted lower amounts of N per ha per year compared to the 100% perennial ryegrass-white clover pastures, but again not by a large amount when it was expressed in N leached/kg LW gain (19.2 g N leached/kg LW gain vs 18.9 g N leached/kg LW gain, Table 3).

Discussion

Modelling limitations

Although the modelling excerise allowed us to analyse major parameters concerning heifer growth and N leaching from calves/heifer production systems, caution is needed to interpret the results. (1). If there are any breed differences in energy requirements or efficiency that are independent of differences in LW and growth rate, these are not accommodated by the use of a generic heifer feed energy requirement equation based on feeding standards in the current study. (2). The urinary N concentration and UN prediction equations were developed based on heifers grazing plantain/ryegrass-white clover pasture studies conducted in Canterbury under irrigation, which is regional specific. Caution is needed to use these equations in any future studies when heifers are grazing in different environment.

Stock number

Calves with a higher target LW had higher ME requirements (AFRC 1995) and therefore this caused the total number of animals on farm to decline (i.e. lower stocking rate) when feed supply was similar across target LW treatments. Within the same target LW, as plantain became a larger part of the diet, feed supply increased due to higher growth rate from plantain than perennial ryegrass-white clover. This allowed more stock to be carried on farm with plantain groups than 100% perennial ryegrass-white clover group.

Animal N intake and LW gain

In spring, as the plantain proportion increased, the seasonal N intake increased by 9%. This was mainly due to the N concentration in plantain being 11% higher than perennial ryegrass-white clover. In contrast to this, in summer, as the plantain proportion increased, the seasonal N intake declined by 8%, attributed to the reduced plantain N concentration than perennial ryegrass-white clover. Across the whole season, as the proportion of plantain increased, total farm N intake increased from 218 kg N/ha/year (for 0% plantain) to 267 kg N/ha/year (for 75% plantain). This was most likely due to an increased feed supply and DMI from feeding plantain to animals, rather than the variations in feed N concentration across forage treatment. In this study, feed N concentration was sufficient to support the target LW gain of growing calves/heifers, according to the recommendation made by Pacheco and Waghorn (2008), who estimated 2.2% N in the diet is sufficient to sustain growing animal performance. Therefore, the LW gain in this study was mostly driven by ME intake (i.e. ME content × DMI). The total LW gain/ha/year increased from 1028 kg/ha/year (for 0% plantain) to 1254 kg/ha/year (for 75% plantain) reflects the higher feed supply and DMI from feeding plantain, as there was no difference observed in ME content between plantain and perennial ryegrass-white clover (12.2 MJ/kg DM and 12.3 MJ/kg DM, respectively) across the whole year. On a whole farm level, N intake (kg N/ha/year) and total LW gain (kg LW gain/ha/year) were similar between the three target LW treatments. This was attributed to similar feed supply being offered to animals across target LW groups, implying that the operating rules were applied effectively.

N excretion and leaching potential

Averaged over all forage treatments, individual animals with a higher target LW ingested more N, which led to higher UN (g/day/animal). On the other hand because of similar N concentration observed between forage treatments, no difference was found in UN per animal (ranged between 77 and 86 g N/day/animal), which is in agreement with previous study results from heifers fed with plantain (Cheng et al. 2017a). Total N leaching levels were much lower in spring and summer than in autumn and winter, which reflects the seasonal variation in N leaching potential described by Carlton et al. (2016). While individual animals with a higher target LW had higher UN (g/day/animal), N leaching decreased as target LW increased as a result of decreased stock number (i.e. stocking rate). Similarly, the higher stocking rate from plantain feeding than perennial ryegrass-white clover feeding led to an increased annual N leaching value, partly contributed from more urine patch numbers. The increased N leaching with plantain inclusion rate (Table 3) was consistent across target LW groups; it increased by 19.8% (20.7–24.8 kg/ha/yr) with the lowest target LW heifers, 19.8% (19.7–23.6 kg/ha/yr) with the intermediate group and 20% (19–22.8 kg/ha/yr) with the largest animals. This finding is in contrast to most of the speculated outcomes derived from animal studies which don't take into account stocking rate effects on N leaching (Totty et al. 2013; Cheng et al. 2017a; Cheng et al. 2017b). This indicates an animal production system approach to understand the potential value to use plantain to mitigate N leaching is a crucial aspect in future work. It is important to note that the finding from this study assumed that stocking rate increase as a result of higher feed supply, which is not nessesarily relavent to other production systems. For example, a fixed stocking rate system can harvest suplus feed to sell as silage to make profit, without increasing stock numbers and N leaching. Further, this study used data collected from a single experimental site in Canterbury, which had a higher annual average forage growth rate for plantain than perennial ryegrass-white clover. The outcome of the modelling may be different, if plantain has a lower or equal forage growth rate compared with perennial ryegrass-white clover. Interestingly, when looking at system N efficiency expressed in N leached/kg LW gain comparing between the pasture mixtures, the high plantain diet had similar N efficiency to the perennial ryegrass pasture dominant diet. This is supported by similar animal N use efficiency (indicated by plasma urea N) measured in a recent study with heifers fed on different proportions of plantain in the diet (Cheng et al. 2017b).

Conclusion

The higher feed supply and DMI from feeding plantain in this study compared with feeding perennial ryegrass-white clover increased animal LW gain, but on a whole farm level, total LW gain was similar between the three target LW treatments. Individual animals with a higher target LW had higher UN. On the other hand, between forage treatments there was no difference found in UN per animal. Stocking rate was a major driver of N leaching in this study, which resulted in a decreased N leaching with increased target LW groups. Similarly, the higher stocking rate from plantain feeding led to an increased annual N leaching value. At a production system level, N efficiency expressed in N leached/kg LW gain was smilar across treatment groups.

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